

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Write a detailed explanation of the manufacturing of ammonia, including the chemical reactions, key raw materials, and optimal conditions for ammonia production.
- Q.24 Explain the refining and distillation processes, focusing on Atmospheric Distillation Unit (ADU) and vacuum Distillation Unit (VDU). Accompany your explanation with a diagram illustrating the key components and flow of these units.
- Q.25 Discuss a detailed process description of Sugar production, with a neat diagram.

No. of Printed Pages : 4

220631

Roll No.

3rd Sem / Chemical (Pulp & Paper)

Subject : Process Industries

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 In N, P, K fertilizers, what does "N" represent?
- a) Nitrogen b) Nickel
c) Neodymium d) Nitrate
- Q.2 What is the main product of the sugar industry?
- a) Molasses b) Ethanol
c) Sucrose d) Glucose
- Q.3 What is the primary raw material in the production of Portland cement?
- a) Limestone b) Clay
c) Gypsum d) Silica

Q.4 What is the pretreatment in petroleum refining?

- a) Cracking b) Desalting
- c) Dewaxing d) Desulfurization

Q.5 In polymerization, what is the role of catalyst?

- a) Initiates the reaction
- b) Inhibits the reaction
- c) Increases pressure
- d) Reduces temperature

Q.6 What does the flash point of petroleum indicate?

- a) Pour point b) Boiling point
- c) Freezing point d) Ignition temperature

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 What is petroleum refining?

Q.8 Write raw material used in cement industry.

Q.9 What is cloud point?

Q.10 Write chemical formula of Ammonia?

Q.11 Define Dehydration?

Q.12 Name any two micronutrients.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Difference between Octane number and cetane number?

Q.14 Explain the manufacturing of sugar with flow sheet

Q.15 Explain the concept of pretreatment of crude oil.

Q.16 Discuss about the manufacturing of urea with chemical reaction, flow sheet and process description.

Q.17 Discuss in detailed Thermal and Catalytic Reforming in petroleum refining.

Q.18 What is fertilizer, explain about its different types.

Q.19 Discuss major engg. Problems associated with Cement industry.

Q.20 Write the difference between setting process and filtration.

Q.21 Explain about the Significance of Petroleum properties.

Q.22 Explain the origin, composition, and classification of crude oil, emphasizing their importance in the refining process.